

# European day-ahead electricity market clearing model



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## ABSTRACT

Following the latest directives, the European Union promoted the integration of local/national electricity markets into one multi-area market, thus allowing the optimal exploitation of the scarce interconnection transmission capacity. In this paper, the authors present a mathematical formulation that explicitly incorporates nearly all market products and transmission constraints currently appearing in the European Power Exchanges (PXs). The proposed clearing model incorporates an optimization problem, formulated as a mixed integer linear programming model, and is solved within an iterative algorithm for the handling of paradoxically accepted block and minimum income orders.

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## 1. Introduction

### 1.1. Background and aim

European power exchanges (PXs) constitute marketplaces where two-sided supply (portfolio- or unit-based) and demand orders are effectively cleared in the presence of critical line congestion constraints, and settled in a uniform, marginal price per bidding area. During the last decade, the European Commission has prioritized the creation of a single internal electricity market, aiming at the effective trading of electricity products in different bidding areas and time periods [1].

In this direction, important steps have been achieved lately with the creation of a pan-European day-ahead electricity market. In June 2012 and following the publication of the Network Codes on capacity allocation and congestion management [2] by ENTSO-E (European Network of Transmission System Operators for electricity), seven PXs of the European region launched the Price Coupling of Regions (PCR) initiative [3], which aimed at the integration of the European day-ahead electricity markets. The successful launching of the PCR market clearing took place on February 4th 2014 [4], effectively coupling in this way the CWE (Central-Western European) region and the Nord Pool Spot market. Further important steps in the creation of the pan-European day-ahead

electricity market were accomplished with the implicit auctioning of the Spain–France border (13th May 2014 [5]) and the successful coupling of the Italian market with the other European markets on February 24th 2015 [6]. In this way, the integrated European electricity market currently clears on average daily basis more than 4 TWh of electric power. Finally, on May 20th 2015, another important step towards the efficient European electricity market integration was achieved, since the Flow-Based (FB) approach was confirmed to go-live in the CWE region [7].

### 1.2. Literature review

The explicit modeling of European PXs has been an extremely timely and interesting topic. In this direction, various researchers have presented formulations that model and handle most of the trading products that are available in European PXs.

One basic characteristic of most European day-ahead electricity markets lies in the inherent non-convex character of such markets. Most PXs allow the submission of multi-period energy products, which emulate some of the unit technical and operational characteristics and allow market participants to internalize all different costs (variable and fixed costs) in the same offer. The most common types of such orders are the block, linked block and flexible hourly orders; due to their “fill-or-kill” nature they require the use of binary variables for their explicit modeling. The explicit modeling of indivisible block (regular, linked and convertible) and flexible hourly orders, which constitute the basic multi-period products traded in European PXs, has been presented in

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## Nomenclature

### Indices and Sets

|            |                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $t(T)$     | index (set) of trading periods within the trading day (typically, the period is 1 h)                                                                                                                                                                                                                                                                                                                         |
| $a(A)$     | index (set) of bidding areas                                                                                                                                                                                                                                                                                                                                                                                 |
| $b(B)$     | index (set) of supply offers' and demand bids' steps                                                                                                                                                                                                                                                                                                                                                         |
| $d(D)$     | index (set) of demand bids, $D_a \subseteq D$ is the subset of demand bids submitted to bidding area $a$                                                                                                                                                                                                                                                                                                     |
| $eg(EG)$   | index (set) of exclusive groups                                                                                                                                                                                                                                                                                                                                                                              |
| $l(L)$     | index (set) of interconnections, where $L_{ac} \cup L_{dc} = L$ are the subsets of AC and DC interconnections and $L_{ls} \subseteq L$ is the subset of interconnections belonging to the same line set                                                                                                                                                                                                      |
| $s(S)$     | Index (set) of supply, $S_a \subseteq S$ is the subset of supply offers submitted to bidding area $a$ , $S_{lg} \subseteq S$ the subset of supply offers that are subject to a Load Gradient Condition, $S_{mic} \subseteq S$ the subset of supply offers that are subject to a Minimum Income Condition and $S_{ss} \subseteq S$ the subset of supply offers that are subject to a Scheduled Stop Condition |
| $bo(BO)$   | index (set) of block orders (profile or regular block orders), $BO_a \subseteq BO$ is the subset of block orders submitted to bidding area $a$                                                                                                                                                                                                                                                               |
| $lbo(LBO)$ | index (set) of linked block orders, $LBO_a \subseteq LBO$ is the subset of linked block orders submitted to bidding area $a$                                                                                                                                                                                                                                                                                 |
| $fho(FHO)$ | index (set) of flexible hourly orders, $FHO_a \subseteq FHO$ is the subset of flexible hourly orders submitted to bidding area $a$                                                                                                                                                                                                                                                                           |
| $ls(LS)$   | index (set) of line sets                                                                                                                                                                                                                                                                                                                                                                                     |

### Parameters

|                                |                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $P_{db}^t, Q_{db}^t$           | price-quantity pair of step $b$ of the hourly priced demand bid $d$ in trading period $t$ , in €/MWh and MWh, respectively                                               |
| $P_{sb}^t, Q_{sb}^t$           | price-quantity pair of step $b$ of the hourly priced energy offer $s$ in trading period $t$ , in €/MWh and MWh, respectively                                             |
| $P_{bo}, Q_{bo}^t$             | price-quantity pair of block order $bo$ , in €/MWh and MWh, respectively; in the case of a profile block order, the quantity may be different in each trading period $t$ |
| $P_{fho}, Q_{fho}$             | price-quantity pair of flexible hourly order $fho$ , in €/MWh and MWh, respectively                                                                                      |
| $F_{lt}^{\min}(F_{lt}^{\max})$ | minimum (maximum) available capacity (also called Remaining Available Margin in the flow-based model) of interconnection $l$ in trading period $t$ , in MW               |
| $R_{bo}^{\min}$                | Minimum Acceptance Ratio of block order $bo$ , in p.u.                                                                                                                   |
| $PTDF_{aa'}^{lt}$              | power transfer distribution factor (PTDF) of AC interconnection $l$ for an energy transfer (exchange) between bidding areas $a$ and $a'$ in trading period $t$ , in p.u. |
| $A_{bo}^{lbo}$                 | linked block order $lbo$ to block order $bo$ incidence matrix                                                                                                            |
| $A_{bo}^{eg}$                  | block order $bo$ to exclusive group $eg$ incidence matrix                                                                                                                |
| $\tau_{lt}$                    | flow tariff for the utilization of DC interconnection $l$ in trading period $t$ , in €/MWh                                                                               |
| $G_s^{dn}(G_s^{up})$           | decrease (increase) gradient of the supply order $s$ , which is subject to the Load Gradient Condition, in MW/min                                                        |
| $B_{sbt}^{ss}$                 | ancillary parameter denoting the activation of the Scheduled Stop Condition, if equal to 1                                                                               |

|                                  |                                                                                                                                                                                                                                                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $B_s^{mic}$                      | ancillary parameter denoting the number of available opportunities a supply offer $s$ has, in order to fulfil its Minimum Income Condition                                                                                                                                                                |
| $\chi_l^a(\varphi_l^a)$          | parameter denoting that DC interconnection $l$ begins (ends) from (to) bidding area $a$ , if equal to 1; otherwise it is equal to 0                                                                                                                                                                       |
| $\lambda_l^a$                    | loss factor of DC interconnection $l$ when power is transferred to bidding area $a$ , in %                                                                                                                                                                                                                |
| $R_{at}^{H,dn}(R_{at}^{H,up})$   | hourly ramp-down (ramp-up) limit of bidding area's $a$ net position, in trading period $t$ , in MW/h; the term "net position ramping" refers to the hourly difference in the net position of a bidding area between two consecutive trading periods of the trading day.                                   |
| $P_{at}^{H,ini}$                 | parameter denoting the initial net position of bidding area $a$ in trading period $t$ ; it is equal to the net position of bidding area $a$ during the last hour of the previous trading day for the first trading period ( $t=1$ ) and zero for all other trading periods ( $t \in \{2-24\}$ ), in MW    |
| $R_a^{D,dn}(R_a^{D,up})$         | daily ramp-down (ramp-up) limit of bidding area's $a$ net position, in MW (per day)                                                                                                                                                                                                                       |
| $P_a^{D,ini}$                    | total net position of bidding area $a$ during the previous trading day, in MW                                                                                                                                                                                                                             |
| $R_{lt}^{H,dn}(R_{lt}^{H,up})$   | hourly ramp-down (ramp-up) limit of interconnection $l$ power flow, in trading period $t$ , in MW/h                                                                                                                                                                                                       |
| $F_{lt}^{H,ini}$                 | parameter denoting the initial power flow in interconnection $l$ in trading period $t$ ; it is equal to the power flow of interconnection $l$ during the last hour of the previous trading day for the first trading period ( $t=1$ ), and zero for all other trading periods ( $t \in \{2-24\}$ ), in MW |
| $R_{ls}^{LS,dn}(R_{ls}^{LS,up})$ | ramp-down (ramp-up) limit of line set $ls$ total power flow, in MW/h                                                                                                                                                                                                                                      |
| $F_{ls,t}^{LS,ini}$              | parameter denoting the initial total power flow in line set $ls$ in trading period $t$ ; it is equal to the total power flow of line set $ls$ during the last hour of the previous trading day for the first trading period ( $t=1$ ) and zero for all other trading periods ( $t \in \{2-24\}$ ), in MW  |

### Variables

#### (i) Positive Variables

|                      |                                                                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $p_{at}$             | net position of bidding area $a$ in trading period $t$ , in MWh; the term "net position" refers to the net injection of a bidding area, namely total cleared production minus total cleared demand. |
| $f_{lt}$             | power flow in interconnection $l$ in trading period $t$ , in MW                                                                                                                                     |
| $e_{aa'}^t$          | bilateral exchange between bidding area $a$ and bidding area $a'$ , in trading period $t$ , in MWh                                                                                                  |
| $x_{sb}^t(x_{db}^t)$ | acceptance ratio of step $b$ of supply (demand) entity's $s$ priced offer (bid) in trading period $t$                                                                                               |
| $x_{bo}$             | acceptance ratio of block order $bo$                                                                                                                                                                |
| $f_{lt}^+, f_{lt}^-$ | positive variables utilized in the power flow variable decomposition schema for DC inter-connection $l$ in trading period $t$ , in MW                                                               |

#### (ii) Binary Variables

|             |                                                                      |
|-------------|----------------------------------------------------------------------|
| $u_{bo}$    | clearing status of block order $bo$                                  |
| $u_{fho}^t$ | clearing status of flexible hourly order $fho$ in trading period $t$ |

references [8–20], whereas their effective clearing has been handled by either non-linear pricing [8,9] and Lagrangian Relaxation [10] techniques, bilevel and primal-dual formulations [11–14] or through the explicit modeling of Market Clearing Prices (MCPs) as problem variables [15–17].

More specifically, O'Neill et al. present in [8] the modeling of a single area European electricity day-ahead market with indivisible block orders and propose the (common in the U.S. markets) uplift payment method for the compensation of participants whose orders are paradoxically accepted even though they are “out-of-the-money”.<sup>1</sup> The same compensation methodology is also presented in [9], where van Vyve compares modeling-wise the U.S. and European day-ahead electricity markets. In this working paper, it is concluded that the application of the Strict Linear Pricing mechanism in the multi-area European day-ahead electricity market leads to the appearance of Paradoxically Accepted “out-of-the-money” and Paradoxically Rejected “in-the-money” (PRO) orders, which stems from the non-convex nature of the underlying optimization problem.

In a different direction, Basagoiti et al. present in [10] a Lagrangian Relaxation based method for the decentralized market integration of two PXs with block orders, whereas Zak et al. [11] formulate a bilevel optimization problem for the clearing of a single area day-ahead electricity market with block orders, where explicit positive welfare constraints are incorporated in the optimization process, for the handling of paradoxically accepted orders.

Reference [12] constitutes the first work that explicitly deals with the efficient handling of Paradoxically Rejected Orders, by providing a based on the strong duality theory primal-dual formulation for the minimization of the total opportunity costs of the aforementioned type of orders. The same authors present in [13] a similar to [12] MIP-based primal-dual formulation for the efficient modelling of the European day-ahead electricity market, which avoids the incorporation of additional auxiliary continuous or binary variables. Moreover, the authors propose a Benders-like decomposition algorithm for the efficient handling of instances of the branch-and-bound procedure of the MIP solution process where no valid European prices are recognized. In addition, in [14] the aforementioned mathematical optimization model [13] is enhanced by providing a heuristic methodology for the handling of the traded volume maximization problem.

The explicit modeling of Market Clearing Prices as problem variables has been presented in [15–17]. Reference [15] constitutes one of the first papers that discusses the appearance and implications of paradoxically accepted orders in European combinatorial auctions. The handling of PAO is performed through the incorporation of pseudo-variables in the optimization problem, for the explicit modeling of MCPs, whereas additional constraints are imposed in the model in order to ensure that energy always flows from a lower MCP area to a higher one. Additionally, Meeus et al. [16] also use in their formulation explicit variables for the modeling of MCPs and perform an extensive analysis on the effect of block orders to the total computational time and on the effect of the block orders' number and maximum size to the appearance of Paradoxically Rejected Orders. Finally, Derinkuyu [17] also uses the aforementioned modeling methodology and proposes methods based on aggregation techniques and variable elimination for the clearing of the Turkish day-ahead electricity market within acceptable timeframes.

Furthermore, iterative approaches, similar to the ones used by European PXs for the effective clearing of day-ahead

electricity markets, have been proposed by Martin et al. [18] and Biskas et al. [19] in order to clear the pan-European multi-regional day-ahead electricity market, whereas a Mixed Complementarity Problem is used by Vlachos and Biskas in [20], for the modeling and clearing of two new (proposed) market orders, namely the adjustable (divisible) and joint demand profile blocks. In [18], Martin et al. present a large-scale mathematical program with equilibrium constraints (MPEC) for the modeling of the pan-European day-ahead electricity market. For the solution of this complex optimization problem, the authors decompose the original problem into a master welfare maximization problem and a pricing subproblem and solve the decomposed problem with an exact and a heuristic iterative algorithm. An iterative algorithm is also proposed by Biskas et al. for the handling of PAO, which investigate in [19] the coupling of European day-ahead electricity markets that are organized under completely different regulatory frameworks, namely Power Exchanges and unit-commitment based Power Pools.

Moreover, the existence of special market clearing conditions, such as the Minimum Income Condition (MIC) in the Iberian market [21] (aiming at the provision of adequate revenues to producers) or the existence of postage-stamp charges for the consumers in the Italian market (all consumers pay the “National Single Price” – “Prezzo Unico Nazionale” (PUN) – calculated as the weighted average zonal price [22]), require the utilization of iterative methodologies and further complicate the effective clearing of the day-ahead market. The explicit modeling of MIC and PUN orders has been presented in references [14,23,24]. In [23], a Mixed Complementarity formulation is used in order to efficiently model the non-linear clearing condition of PUN orders appearing in the Italian peninsula, whereas bilevel and primal-dual formulations (based on [13]) are presented in [14,24], respectively, that enable the incorporation in the optimization problem of positive revenue constraints that explicitly model the MIC clearing condition.

Finally, it should be noted that most of the aforementioned publications use the ATC-based<sup>2</sup> network representation [9–10,12–15,18], whereas only Martin et al. [18] deal explicitly with the enforcement of line power flow ramping constraints in the optimization problem.

Table 1 presents an analytical comparison between the aforementioned and the model proposed in the following sections.

### 1.3. Clearing model – main contributions

The main contribution of this paper is the explicit modeling (exact problem formulation and iterative algorithm) of the bulk of the network and market constraints currently appearing in European PXs. Moreover, the following specific issues constitute significant contributions of this paper, with regard to the already existing literature:

- i. The derivation of the market clearing prices (MCPs), which are calculated as a function of the Lagrange multipliers of the problem constraints. Such prices are not calculated based solely on the Lagrange multiplier of the power balance equation (as is the common approach in such market clearing models), but depend also on additional terms, which are related to the hourly and daily net position constraints.
- ii. The modelling of profile block orders and block orders in exclusive groups. Such modelling is presented for the first time, since most research works in the literature handle mainly the regular block, linked and flexible hourly orders.

<sup>1</sup> The appearance of Paradoxically Accepted Orders (PAO), namely indivisible orders that are accepted even though they entail a negative welfare to the participant that submits them, in European day-ahead electricity markets is analytically discussed in Section 4.1.

<sup>2</sup> ATC stands for Available Transfer Capacity.

**Table 1**

Comparison between relevant features of works presented in the literature and the herein proposed model.

| Ref        | Model     | Orders |              |                 |                  | PUN Orders | MIC Orders | Network Modeling | Ramping |      | Number of areas  |
|------------|-----------|--------|--------------|-----------------|------------------|------------|------------|------------------|---------|------|------------------|
|            |           | Block  | Linked block | Flexible hourly | Other types      |            |            |                  | Line    | Area |                  |
| [8]        | MILP      | Yes    | No           | No              | No               | No         | No         | No               | –       | –    | –                |
| [9]        | MILP      | Yes    | Yes          | Yes             | No               | No         | No         | ATC              | No      | No   | N/S <sup>d</sup> |
| [10]       | MILP      | Yes    | No           | No              | No               | No         | No         | ATC              | No      | No   | 2                |
| [11]       | MILP      | Yes    | No           | No              | No               | No         | No         | No               | –       | –    | –                |
| [12]       | MILP      | Yes    | No           | No              | No               | No         | No         | ATC              | No      | No   | 4                |
| [13]       | MILP/MIQP | Yes    | No           | No              | No               | No         | No         | ATC & PTDF       | No      | No   | N/S              |
| [14]       | MILP      | Yes    | No           | No              | No               | No         | Yes        | ATC              | No      | No   | 4                |
| [15]       | MILP      | Yes    | No           | No              | No               | No         | No         | ATC              | No      | No   | 12               |
| [16]       | MILP      | Yes    | No           | No              | No               | No         | No         | No               | –       | –    | –                |
| [17]       | MILP      | Yes    | Yes          | Yes             | No               | No         | No         | No               | –       | –    | –                |
| [18]       | MILP      | Yes    | Yes          | Yes             | No               | No         | No         | ATC              | Yes     | No   | 10               |
| [19]       | MILP      | Yes    | Yes          | Yes             | Yes <sup>a</sup> | No         | No         | PTDF             | No      | No   | 42               |
| [20]       | MCP       | No     | No           | No              | Yes <sup>b</sup> | No         | No         | PTDF             | No      | No   | 73               |
| [23]       | MCP       | No     | No           | No              | No               | Yes        | No         | PTDF             | No      | No   | 5/73             |
| [24]       | MILP      | No     | No           | No              | No               | No         | Yes        | PTDF             | No      | No   | 5                |
| This paper | MILP      | Yes    | Yes          | Yes             | Yes <sup>c</sup> | No         | Yes        | PTDF             | Yes     | Yes  | 42               |

<sup>a</sup> Convertible block orders.<sup>b</sup> Adjustable and joint demand.<sup>c</sup> Exclusive group of block orders.<sup>d</sup> N/S acronym stands for not specified.

- iii. The modelling of the line set power flow and (daily/hourly) net position ramping constraints, which are presented for the first time in the existing literature. Additionally, their impact on the formation of the finally attained MCPs is revealed and analyzed.
- iv. The modelling of MIC orders through the use of certain parameters ( $B_s^{mic}$ ) that prevent the acceptance of complex orders that shall most probably not satisfy their minimum income condition, but provide more clearing opportunities to complex orders that are close to fulfill their minimum income condition in the current iteration of the solution process, and thus they may fulfill their minimum income condition in the following iterations. Such handling of the clearing condition of MIC orders renders the use of binary variables to simulate MIC orders unnecessary in herein described model. The use of parameters  $B_s^{mic}$  is described analytically in Section 5.2 of this paper.

The herein proposed clearing model embodies an iterative algorithm for handling the paradoxically accepted block and minimum income orders. In the following sections, the main building concepts and functionality of the proposed clearing model are presented, along with the iterative algorithm used for the effective clearing of the examined day-ahead electricity market. Moreover, the presented market clearing model is evaluated using a real-size test case, and the presented results reveal important insights about the effect of network and market constraints in the attained volumes and prices.

## 2. Transmission network and market orders appearing in European PXs

In the following subsections, the transmission network and market orders appearing in most European PXs are briefly presented.

### 2.1. Transmission network

#### 2.1.1. Power flow ramping constraints

Under this specific set of constraints, the power flow in all transmission lines that are subject to ramping constraints in each trading period  $t$  depends on the respective power flow during the previous trading period  $t - 1$ .

Power flow ramping limitations were traditionally imposed on the HVDC (High Voltage Direct Current) interconnections of the Northern European power system due to the fact that HVDC

interconnections have higher actual ramping capabilities than that of generation and consumption units [25]. As a consequence, Nordic Transmission System Operators (TSOs) enforced hourly ramping limitations (lower ramping limits), in order to avoid frequent or large changes in the DC interconnections' power flows, which gravely influenced the system frequency and required the reservation of large amounts of frequency-related ancillary services for the efficient handling of the system imbalances and frequency fluctuations [26]. Moreover, the need for imposing ramp rates on DC interconnectors has also been highlighted by ENTSO-E on the Requirements Outlines of the "Network code for HVDC connections and DC-connected power market modules" [27], where it is pointed out that TSOs should control the DC ramp rates, in order to ensure the secure system operation, in case of "exhaustion of reserves", due to the high ramping capability of DC interconnectors.

Additionally, power flow ramping constraints may be imposed on a set of lines, as was the real-world case of the SE3-DK1 and NO2-DK1 interconnectors (such a constraint was incorporated in the market clearing due to "internal grid limitations" of northern Denmark [26]).

#### 2.1.2. Net position ramping constraints

Ramping limits may be imposed not only in transmission lines but also in bidding areas' net positions. In this case, ramping limits may be hourly, where a bidding area's net position in trading period  $t$  depends on its net position in the previous trading period  $t - 1$ , and/or daily.

Hourly net position ramping constraints were primarily enforced due to the high hourly net position volatility that was observed in the Flow-Based Market Coupling (FB MC) methodology, since the generating units that are located in the examined bidding areas may not be able to respond to such high changes of the cleared net positions [28]. Moreover, voltage and dynamic stability issues, which are not taken into consideration during the computation of the Flow Reliability Margin parameter of the FB MC method, require the introduction of additional import/export limitations, in order to ensure the secure grid operation [29]. For the aforementioned reasons, CWE TSOs introduced additional hourly ramping constraints in the market clearing process, that limit the net position change.

Finally, the daily net position ramping constraint is incorporated practically to limit the amount of reserve capacity used during the day [22].

### 2.1.3. Tariffs and losses

Owners or operators of DC interconnections may impose tariffs for each transferred MWh (in €/MWh). Moreover, in the ATC model, transmission lines between the interconnected bidding areas may be subject to losses. In such case, the energy received in the importing area is less than the energy delivered in the exporting area.

### 2.2. Market orders in European power exchanges

European PXs allow the submission of the following types of market orders [22]:

- Step-wise or piecewise linear hourly supply and demand orders:* for simplicity reasons, in this paper only stepwise orders have been taken into consideration; the incorporation of piecewise hourly orders is straightforward.
- Complex Orders:* stepwise supply offers with special clearing rules, comprising a Load Gradient Condition, a Minimum Income Condition or a Scheduled Stop Condition. The latter facilitates the power producer to smoothly shut down a power plant that is not scheduled to operate during the trading day, due to the non-fulfillment of its Minimum Income Condition [21]. It should be noted that a supply offer may be subject to all complex conditions simultaneously.
- Block Orders:* A basic characteristic of most central European PXs is the fact that they allow market participants to submit multi-period energy products, which are used for modeling some of the generating units' technical characteristics or for hedging the risk associated with the forecasting of the volatile day-ahead MCPs. These energy products are collectively called block orders, and comprise regular block orders, profile block orders (allowing the partial acceptance of the order), linked block orders (in which the acceptance ratio of the parent block should always be greater than that of the child), block orders in an exclusive group and flexible hourly orders (cleared at most at one trading period that maximizes the total social welfare).
- Merit and PUN Orders:* Merit orders can be both supply hourly offers and demand hourly bids, where a specific merit order number is associated with each respective order, to provide the order of clearing in case of same offer prices by various Merit orders.

PUN orders are not cleared at the MCP of the bidding area that they are submitted, but at the PUN price ("Prezzo Unico Nazionale" price). This specific market clearing condition is enforceable only in the Italian region (consisting of six bidding areas), where demand bids are cleared at the weighted average MCP of the Italian bidding areas.

It should be noted that, even though the aforementioned Merit and PUN orders are tradable in the European day-ahead markets, they are not part of the mathematical formulation of the model presented in the following section. Their explicit modeling and incorporation in the market clearing algorithm involves special clearing rules and procedures; these shall be investigated in terms of the authors' future research.

## 3. Mathematical formulation

### 3.1. Problem formulation

The overall day-ahead welfare maximization problem is modeled as follows:

#### i. Objective function

$$\text{Max}_{v \in V} w^{\text{tot}} = u^{\text{dem}} - (c^{\text{sup}} + c^{\text{bo}} + c^{\text{fho}}) - c^{\text{cr}} \quad (1)$$

Eq. (1) represents the objective function of the Mixed Integer Linear Programming (MILP) optimization problem, aiming at the maximization of the overall social welfare  $w^{\text{tot}}$ , under the set of primal decision variables  $V = \{u_{bo}, u_{fho}^t, x_{db}^t, x_{sb}^t, x_{bo}, e_{aa}^t, p_{at}, f_{lit}\}$ . The overall social welfare comprises:

- the total load utility function:  $u^{\text{dem}} = \sum_{d \in D} \sum_{b \in B} \sum_{t \in T} [P_{db}^t \cdot Q_{db}^t \cdot x_{db}^t]$
- the total simple hourly offer cost function:  $c^{\text{sup}} = \sum_{s \in S} \sum_{b \in B} \sum_{t \in T} [P_{sb}^t \cdot Q_{sb}^t \cdot x_{sb}^t]$
- the total block offer cost function:  $c^{\text{bo}} = \sum_{bo \in BO} \sum_{t \in T} [P_{bo} \cdot Q_{bo}^t \cdot x_{bo}]$ , and
- the total flexible hourly offer cost function:  $c^{\text{fho}} = \sum_{fho \in FHO} \sum_{t \in T} [P_{fho} \cdot Q_{fho}^t \cdot u_{fho}^t]$

Moreover, the total welfare  $w^{\text{tot}}$  depends on the cleared power flow of the merchant DC interconnections, since an additional term is subtracted from the objective function, denoting the congestion rent (welfare loss)  $c^{\text{cr}}$  due to the related tariffs, which is equal to:

$$c^{\text{cr}} = \sum_{l \in L_{dc}} \sum_{t \in T} [\tau_{lt} \cdot (f_{lt}^+ + f_{lt}^-)]$$

It should be noted that in the herein proposed mathematical formulation it is assumed that  $Q_{bo}^t, Q_{fho} \geq 0$  for all block and flexible hourly supply offers, respectively, whereas  $Q_{bo}^t, Q_{fho} \leq 0$  for all block and flexible hourly demand bids.

The objective function (1) is subject to the following set of constraints. The dual variables (Lagrange multipliers) of the respective primal constraints, used for the derivation of the finally attained Market Clearing Prices, are shown in brackets at the right side of the respective constraints.

#### ii. Order clearing constraints

Constraints (2)–(11) represent the order clearing rules of the optimization problem.

$$x_{sb}^t \leq 1 \quad \forall s \in S, b \in B, t \in T \quad (2)$$

$$x_{db}^t \leq 1 \quad \forall d \in D, b \in B, t \in T \quad (3)$$

$$\begin{aligned} & \sum_{b \in B} [Q_{sb}^t \cdot x_{sb}^t] - \sum_{b \in B} [Q_{sb}^{t-1} \cdot x_{sb}^{t-1}] \\ & \leq 60 \cdot G_s^{\text{up}} \quad \forall s \in S_{lg}, t \in T - \{t_1\} \end{aligned} \quad (4)$$

$$\begin{aligned} & \sum_{b \in B} [Q_{sb}^{t-1} \cdot x_{sb}^{t-1}] - \sum_{b \in B} [Q_{sb}^t \cdot x_{sb}^t] \\ & \leq 60 \cdot G_s^{\text{dn}} \quad \forall s \in S_{lg}, t \in T - \{t_1\} \end{aligned} \quad (5)$$

$$x_{sb}^t \leq B_s^{\text{mic}} + B_{sbt}^{\text{ss}} \quad \forall s \in S_{mic}, b \in B, t \in T \quad (6)$$

Constraints (2) and (3) denote the upper limits for the clearing of the simple hourly supply offers and demand bids, whereas Eqs. (4) and (5) enforce the Load Gradient Condition to the respective supply orders (Load Gradient Orders), for all trading periods except



from the first one [22]. Constraint (6) enforces simultaneously the Minimum Income and Scheduled Stop Conditions. In equation (6), parameter  $B_s^{mic}$  can take only a positive integer value, representing the number of available opportunities that a supply order  $s$  has in order to fulfill its Minimum Income Condition, before being removed from the Order Book. In this way, in case  $B_s^{mic} \geq 1$ , then Eq. (6) is redundant, since the clearing status of the respective supply order is limited from Eq. (2). On the other hand, in case that  $B_s^{mic} = 0$ , meaning that the respective supply order does not have any other opportunities to fulfill its Minimum Income Condition and is thus effectively removed from the Order Book, Eq. (6) forces the clearing status of the supply order to be equal to zero. The Scheduled Stop Condition is enforced through parameter  $B_{sbt}^{ss}$ , which is equal to zero for all supply offers, steps and trading periods, unless supply order  $s$  is subject to the Scheduled Stop Condition ( $s \in S_{ss}$ ). In this case, the aforementioned parameter is equal to 1 only for the first step and the first three trading periods. Thus, even if the respective order fails to fulfill its Minimum Income Condition ( $B_s^{mic} = 0$ ), the first sub-hourly order (step) is subject to clearing for the first three trading periods, activating in this way the Scheduled Stop Condition.

$$R_{bo}^{\min} \cdot u_{bo} \leq x_{bo} \quad \forall bo \in BO \quad (7)$$

$$x_{bo} \leq u_{bo} \quad \forall bo \in BO \quad (8)$$

$$x_{lbo} \leq \sum_{bo \in BO} [A_{bo}^{lbo} \cdot x_{bo}] \quad \forall lbo \in LBO \quad (9)$$

$$\sum_{bo \in BO} [A_{bo}^{eg} \cdot x_{bo}] \leq 1 \quad \forall eg \in EG \quad (10)$$

$$\sum_{t \in T} u_{fho}^t \leq 1 \quad \forall fho \in FHO \quad (11)$$

Constraints (7) and (8) enforce that the clearing status of a profile block order should always be zero or between its minimum and maximum acceptance ratio (the maximum acceptance ratio is equal to 1). In case of a regular block order  $R_{bo}^{\min} = 1$  and thus constraints (7) and (8) degenerate into the classical “fill-or-kill” constraints. Constraint (9) denotes the acceptance condition of a linked block order  $lbo$ , where (according to [22]) the acceptance ratio of a parent block  $bo$  should always be equal to or greater than the acceptance ratio of its child block  $lbo$ . Constraint (10) forces the sum of the accepted ratios of block orders belonging to exclusive group  $eg$  to be less than or equal to 1. Note that in the case of regular block orders belonging to an exclusive group, only one block order may be accepted at any given time. Finally, constraint (11) presents the clearing condition of flexible hourly orders.

### iii. Power balance constraints

$$\begin{aligned} p_{at} = & \sum_{s \in S_a} \sum_{b \in B} [Q_{sb}^t \cdot x_{sb}^t] - \sum_{d \in D_a} \sum_{b \in B} [Q_{db}^t \cdot x_{db}^t] + \sum_{bo \in BO_a} [Q_{bo}^t \cdot x_{bo}^t] \\ & + \sum_{fho \in FHO_a} [Q_{fho}^t \cdot u_{fho}^t] \quad \forall a \in A, t \in T \quad [\xi_{at}] \quad (12) \\ & \sum_{a' \in A} [e_{aa'}^t - e_{a'a}^t] + \sum_{l \in L_{dc}} [\lambda_l^a \cdot (f_{lt}^+ - (1 - \lambda_l^a) \cdot f_{lt}^-)] \\ & - \sum_{l \in L_{dc}} [\varphi_l^a \cdot ((1 - \lambda_l^a) \cdot f_{lt}^+ - f_{lt}^-)] = p_{at} \quad \forall a \in A, t \in T \quad [\xi_{at}] \quad (13) \end{aligned}$$

Eq. (12) provides the net position of bidding area  $a$  in trading period  $t$ . Eq. (13) expresses the hourly power balance equation of

each bidding area  $a$ ; the net position of area  $a$  is equal to the bilateral exchanges between this area and all its neighboring areas (with AC and DC interconnections), considering possible losses. It should be noted that in case bidding areas  $a$  and  $a'$  are not connected through an AC interconnection, then  $e_{aa'}^t = 0$ .

### iv. Power flow constraints

$$f_{lt} = \sum_{a \in A} \sum_{a' \in A} [PTDF_{aa'}^{lt} \cdot e_{aa'}^t] \quad \forall l \in L_{AC}, t \in T \quad (14)$$

$$f_{lt} = f_{lt}^+ - f_{lt}^- \quad \forall l \in L, t \in T \quad [\mu_{lt}^{tot}] \quad (15)$$

$$f_{lt} \leq F_{lt}^{\max} \quad \forall l \in L, t \in T \quad [\mu_{lt}^{\max}] \quad (16)$$

$$F_{lt}^{\min} \leq f_{lt} \quad \forall l \in L, t \in T \quad [\mu_{lt}^{\min}] \quad (17)$$

The power flow constraints are modeled through (14)–(17). The optimization model incorporates a DC transmission network configuration (without AC power flow constraints), where Eq. (14) is used for the modeling of the power flows of the AC interconnectors, and the power flow of the DC interconnectors is modeled as a free variable, according to [19]. The calculation of the  $PTDF_{aa'}^{lt}$ , used in (14), requires initially the definition of a reference bidding area  $ref$  for each synchronous area of the European power system, and subsequently the computation of the  $PTDF_{a,ref}^{lt}$ , considering the AC interconnections' admittances [30]. Then,  $PTDF_{aa'}^{lt}$  is given by:

$$PTDF_{aa'}^{lt} = PTDF_{a,ref}^{lt} - PTDF_{a',ref}^{lt} \quad \forall a, a' \in A, l \in L_{AC}, t \in T \quad (18)$$

It should be noted that in case the internal lines' congestion is taken into consideration in the market clearing algorithm, as in the case of CWE region [28], then appropriate Generation Shift Keys (GSKs) should be defined (along with the use of nodal PTDFs) for the computation of the  $PTDF_{aa'}^{lt}$  matrix [31].

Eq. (15) decomposes the free power flow variable into two positive variables, whereas Eqs. (16) and (17) express the upper and lower power flow limits in each trading period  $t$ .

### v. Hourly and daily net position ramping constraints

$$p_{at} - p_{a,t-1} - P_{at}^{H,ini} \leq R_{at}^{H,up} \quad \forall a \in A, t \in T \quad [\beta_{at}^{H,up}] \quad (19)$$

$$p_{a,t-1} - p_{at} + P_{at}^{H,ini} \leq R_{at}^{H,dn} \quad \forall a \in A, t \in T \quad [\beta_{at}^{H,dn}] \quad (20)$$

$$\sum_{t \in T} p_{at} - P_a^{D,ini} \leq R_a^{D,up} \quad \forall a \in A \quad [\beta_a^{D,up}] \quad (21)$$

$$P_a^{D,ini} - \sum_{t \in T} p_{at} \leq R_a^{D,dn} \quad \forall a \in A \quad [\beta_a^{D,dn}] \quad (22)$$

Constraints (19) and (20) describe the hourly net position ramping constraints. Moreover, constraints (21) and (22) denote the daily net position ramping constraints of bidding area  $a$ ; the total net position of area  $a$  depends on the total net injection  $P_a^{D,ini}$  during the previous trading day.

### vi. Hourly individual line and line set power flow ramping constraints

$$f_{lt} - f_{l,t-1} - F_{lt}^{H,ini} \leq R_{lt}^{H,up} \quad \forall l \in L, t \in T \quad [\gamma_{lt}^{H,up}] \quad (23)$$

$$f_{l,t-1} - f_{lt} + F_{lt}^{H,ini} \leq R_{lt}^{H,dn} \quad \forall l \in L, t \in T \quad [\gamma_{lt}^{H,dn}] \quad (24)$$

$$\sum_{l \in L_{ls}} f_{lt} - \sum_{l \in L_{ls}} f_{l,t-1} - F_{ls,t}^{LS,ini} \leq FR_{ls}^{LS,up} \quad \forall ls \in LS, t \in T \quad \left[ \gamma_{ls,t}^{LS,up} \right] \quad (25)$$

$$\sum_{l \in L_{ls}} f_{l,t-1} - \sum_{l \in L_{ls}} f_{lt} + F_{ls,t}^{LS,ini} \leq FR_{ls}^{LS,dn} \quad \forall ls \in LS, t \in T \quad \left[ \gamma_{ls,t}^{LS,dn} \right] \quad (26)$$

Constraints (23) and (24) express the hourly flow ramping constraints on individual lines and are analogous to the hourly net position ramping constraints (19) and (20). Finally, constraints (25) and (26) express the hourly flow ramping constraints on line sets, where the total power flow of all interconnections  $l$  belonging to line set  $LS$  in trading period  $t$  depends on the total power flow of the respective interconnections in trading period  $t - 1$ .

### 3.2. Derivation of market clearing prices

The incorporation of integer (binary) decisions in the market clearing process leads to the formation of market clearing prices that are not easily defined and economically explained [32–34]. As a matter of fact, the modeling of such integer decisions through the usage of binary variables results in non-convex market-clearing problems (formulated as mixed integer linear programming models), which possibly lead to non-zero (positive) duality gaps in the finally attained solution. As a result, the associated dual variables (used in the pricing mechanism) are not as conveniently interpreted as price signals [35], since such prices in general do not result in a competitive equilibrium (in such cases, an equilibrium may be achieved only in the rare occasion where the relaxed – in terms of integer variables – problem has the same solution as the examined MILP problem).

The aforementioned problem is evident both in the U.S. and the European day-ahead electricity markets. In U.S. electricity markets, structured usually as unit commitment-based power pools, the mixed integer linear market clearing problem includes binary variables associated with the start-up/shut-down decisions, various commitment decisions etc., and leads to the formation of Market Clearing Prices, which “usually leave some executed orders out-of-the-money and/or some rejected orders in-the-money” [9]. In order to deal with such undesirable market clearing results, US regulators have introduced uplifts (bid-cost recovery mechanism or “make-whole payments” [36]) in the settlement process, so as to financially compensate all market participants that are at a loss. The same problem is evident also on the European day-ahead electricity markets, which have traditionally adopted the linear pricing scheme (namely, all participants are remunerated with a single price per bidding area and per trading period) [9]. Such pricing scheme stems from the “Network Code on Capacity Allocation and Congestion Management” [2], where it is explicitly stated (Article 46, clause 2) that the developed pan-European Price Coupling Algorithm shall determine “a single Clearing price for each Bidding Zone and Market Time Period”. Such prices may also lead to undesirable clearing results, since (as further explained in the following Sections) submitted indivisible multi-period block orders may be Paradoxically Accepted or Rejected (accepted even though they are out-of-the-money, or rejected even though they are in-the-money, respectively) [19].

In this paper, the final Market Clearing Prices are calculated based on the method presented by O’Neil et al. [34], namely after finding the optimal binary variables, the same (yet continuous) problem is solved (inherently by CPLEX) by fixing the binary variables at their final (optimal) values, and the shadow prices of all market and network constraints are attained. In this way, the hourly

MCPs,  $\pi_{at}$ , of each bidding area  $a$  are computed for each period  $t$  as follows:

$$\pi_{at} = \xi_{at} - (\beta_{at}^{H,up} - \beta_{a,t+1}^{H,up}) + (\beta_{at}^{H,dn} - \beta_{a,t+1}^{H,dn}) - \beta_a^{D,up} + \beta_a^{D,dn} \quad \forall a \in A, t \in T \quad (27)$$

As shown, the final MCPs depend on the Lagrange Multipliers of both the power balance equation and the hourly and daily net position ramping constraints.

It should be noted that, even though in both the U.S. and the European day-ahead markets a MILP model is formulated and solved, the price-setting process is completely different:

- a) In the U.S. day-ahead markets (e.g. PJM, New York ISO, ISO New England), a Security Constrained Unit Commitment (SCUC) is initially performed incorporating DC power flow equations and security constraints (which are ignored in European markets). The SCUC is followed by a Security Constrained Economic Dispatch (SCED), with the full DC network representation along with linear  $N - 1$  security constraints (for a critical contingency set) to account for the system security. After an iterative procedure resolving all possible congestions and  $N - 1$  security issues, an AC power flow is solved to determine the final area interchange values. This SCED procedure defines the day-ahead Locational Marginal Prices (nodal LMPs) that will be applied in the day-ahead settlement for generation and demand, as a function of (a) the energy price, (b) congestion price and (c) marginal or average losses price.
- b) In the European markets, no security constraints are considered, there is no loss price component (a simple DC network representation is used), but there are hourly and daily net position ramping constraints (19)–(22) (which are absent in U.S. markets) that affect the attained zonal prices (cf. nodal in the U.S. markets) through their Lagrange Multipliers, as shown in (27).

## 4. Price-related issues

This section presents the basic characteristics of two price-related issues, which have been taken explicitly into consideration in the herein proposed market clearing algorithm.

### 4.1. Paradoxically accepted orders

The Paradoxically Accepted Orders (PAOs) are block orders that are accepted in the final market clearing solution, even though their acceptance entails a negative welfare to the participant that submits them. The presence of PAOs in the day-ahead European electricity auctions has been identified from all PXs allowing the submission of block products. Their appearance is attributed to the inclusion of indivisible block and flexible hourly orders, whose explicit modeling requires the introduction of binary integer variables in the overall optimization problem [19]. In this way, the optimization problem is formulated as a non-convex Mixed Integer Linear Program. As a result, MCPs that support the market equilibrium cannot be generally defined [9] and “price jumps” appear in the final solution, where an order may be accepted/rejected, contrary to the market signals. The problem of pricing over non-convexities has been extensively studied in the current literature, whereas the aforementioned orders are designated as Paradoxically Accepted/Rejected Orders [19] and are undesirable both for PXs and participants.

Even though both types of orders seem to be problematic, European PXs exclude only the Paradoxically Accepted Orders from the attained market clearing solution, since such transactions entail a negative welfare to the participants submitting them. On the other

hand, a Paradoxically Rejected Order implies a foregone opportunity to the respective market participant in a theoretical level, since economically concise MCPs cannot be obtained in the MILP paradigm [9].

For the efficient handling of the aforementioned problem, most PXs have historically adopted iterative solutions [37–39] aiming at reaching a solution where the finally derived MCPs do not lead to PAOs. Moreover, various researchers have proposed methodologies based on Lagrangian relaxation [10], bi-level programming [11] and primal-dual formulations [12–14], iterative procedures [18–19,40] or the explicit modeling of MCPs as variables of the optimization problem [15–17] for the efficient handling of PAOs. In all cases, the final market clearing solution does not contain any cleared block orders that entail a loss to the respective participant.

It should be noted that in European markets, since the PAOs are excluded/rejected during the solution iterations, there is no need to apply any uplifts to compensate participants for accepted blocks that were out-of-the-money, since there are no such blocks in the final solution.

#### 4.2. Minimum income condition

The Minimum Income Condition order type is tradable only in the Iberian PX. As already stated in Section 2, a market participant may request a submitted supply offer to be cleared, only if the total cleared volumes of its hourly sub-orders provide enough revenue, to cover a pre-defined fixed and a variable term [22]. In case the finally attained MCPs fail to provide the requested revenue, the respective order along with all its hourly sub-orders is fully rejected.

The subject of revenue-constrained market-clearing has been studied in the literature and solutions have been proposed which either incorporate bi-level optimization methods based on marginal pricing [24,41] or primal-dual formulations aiming at finding adequate prices that fulfill the minimum revenue requirements of all participants [14,42]. Contrary to such one-shot optimization solutions, the Iberian day-ahead electricity market is cleared based on an iterative solution algorithm, where ex-post calculations are performed for the identification of generating units which do not fulfill their Minimum Income Condition. Such units are effectively removed from the Order Book in subsequent iterations and the day-ahead market clearing problem is solved

again, until all cleared quantities and prices provide the required revenue [22].

### 5. Solution process

In this paper, the proposed algorithm for the solution of the pan-European day-ahead market clearing problem employs an iterative procedure, as follows:

- Initially, the overall Mixed Integer Linear Programming (MILP) problem is solved and the optimally cleared volumes of all orders are attained.
- The MCPs for each bidding area and trading period are computed using Eq. (27).
- The algorithm performs two consecutive checks for the determination of the existence of i) Paradoxically Accepted Orders (PAOs), and ii) supply offers that do not fulfill their Minimum Income Condition (MIC).
- In case that either one of the above two checks is not satisfied, then “remedial actions” are taken and the algorithm continues with step (a). These remedial actions involve the handling of PAOs and the handling of MIC orders that do not fulfill their Minimum Income Condition requirement (as illustrated with a red dotted line in Fig. 1). Otherwise, the algorithm terminates.

The PAOs and Minimum Income Condition checks, along with their respective remedial actions, are analytically presented in the following paragraphs.

#### 5.1. Paradoxically accepted orders check

The handling of PAOs is performed through an iterative methodology. In each iteration of the algorithm an ex-post computation is performed, where the block and flexible hourly order total welfare,  $W_{bo}$  and  $W_{fho}^t$  respectively, is computed (based on the attained MCPs), as follows:

$$W_{bo} = \sum_{t \in T} [\pi_{at} - P_{bo}] \cdot Q_{bo}^t \cdot \bar{x}_{bo} \quad \forall bo \in BO_a \quad (28)$$

$$W_{fho} = \sum_{t \in T} [(\pi_{at} - P_{fho}) \cdot Q_{fho} \cdot \bar{u}_{fho}^t] \quad \forall fho \in FHO_a \quad (29)$$

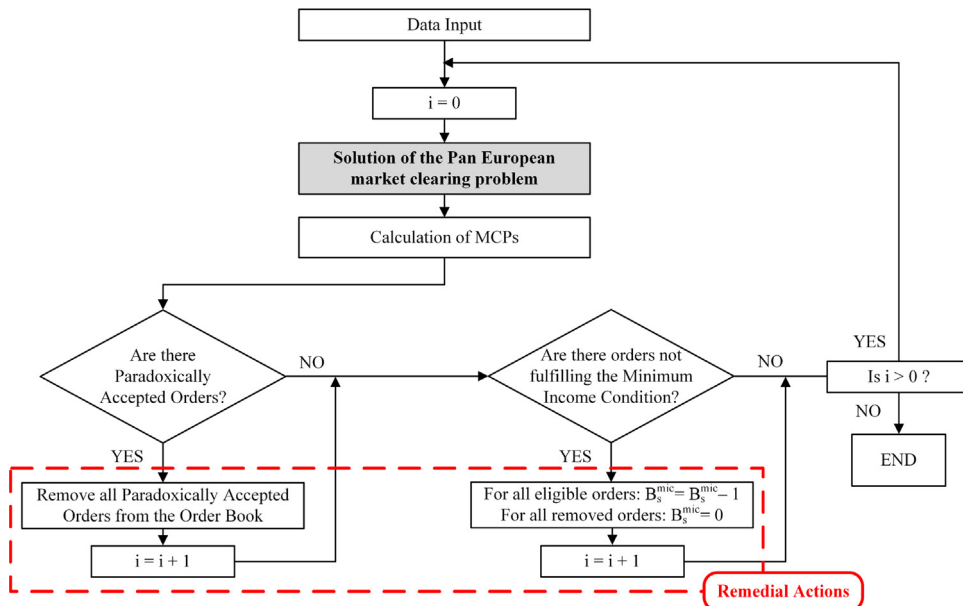


Fig. 1. Schematic illustration of the iterative algorithm.



where  $\bar{x}_{bo}$  and  $\bar{u}_{fho}^t$  are the optimal values of the  $x_{bo}$  and  $u_{fho}^t$  variables, derived by the problem solution at the current iteration of the algorithm. If the attained welfare is negative, then the respective Paradoxically Accepted Block and Flexible Hourly Orders are effectively removed from the Order Book. It should be noted that, following the rules currently appearing in European PXs [22], a negative welfare incurring block order may not be designated as paradoxically accepted, if the welfare of its child linked block order (if such exists) is enough to cover the losses of the parent block.

### 5.2. Minimum income condition check

Additionally, the Minimum Income Condition check is performed. During this check, the attained market revenue  $V_s^{mar}$  of each MIC supply offer at the current iteration is initially calculated by:

$$V_s^{mar} = \sum_{t \in T} [\pi_{at} \cdot Q_{st}^{tot}] \quad \forall s \in S_{mic,a} \quad (30)$$

where  $S_{mic,a} \subseteq S_a$  constitutes the subset of MIC orders submitted to bidding area  $a$ , whereas  $Q_{st}^{tot}$  is the total cleared quantity of the MIC supply order  $s$  in trading period  $t$ . The market revenue is compared to the requested revenue (according to the MIC order terms submitted by the participant),  $V_s^{req}$ , which comprises a variable term  $VT_s$  and a fixed term  $FT_s$  and is calculated as follows:

$$V_s^{req} = VT_s \cdot \sum_{t \in T} Q_{st}^{tot} + FT_s \quad \forall s \in S_{mic} \quad (31)$$

In case a MIC supply offer  $s$  fails to fulfill its Minimum Income Condition ( $V_s^{mar} \leq V_s^{req}$ ), then the  $B_s^{mic}$  parameter in Eq. (6) (denoting the PX-defined number of available opportunities that MIC orders have, in order to fulfill their Minimum Income Condition) takes a value equal to zero and as a result all associated sub-hourly priced energy offers are effectively removed from the Order Book.

It should be noted that during the iterations another opportunity is given to MIC orders, which are very close to fulfill their Minimum Income Condition. In this case, if the attained market revenue  $V_s^{mar}$  is close to the required revenue  $V_s^{req}$ , meaning that there is a high probability that the respective order will be able to fulfill its Minimum Income Condition in the next iteration(s) of the market clearing algorithm, then the value of parameter  $B_s^{mic}$  is reduced by 1, denoting in this way that another opportunity is given to the respective MIC offer, for fulfilling its Minimum Income Condition. Such orders will be hereinafter referred to as “eligible” MIC orders. The aforementioned check is performed by comparing the  $[V_s^{req} - V_s^{mar}] / V_s^{req}$  ratio with a predefined parameter  $X$  (e.g. 20%).

The aforementioned procedure continues until all available opportunities are depleted, and the respective MIC order is eventually removed from the Order Book.

The aforementioned solution algorithm is graphically presented in Fig. 1.

It should be noted that the aforementioned price-related issues are addressed in the proposed market clearing algorithm in an ex-post procedure. As previously stated, after a solution of the master problem has been obtained, the algorithm performs the two ex-post checks in order to determine the existence of paradoxically accepted orders and orders that do not fulfill their minimum income condition. In case these price-related issues appear in the current solution, then the remedial actions described in this Section are performed and the master problem is re-initialized afterwards. As a result, the order of realization of the two ex-post checks does not affect the final market clearing outcome; performing them in a different order would lead to exactly the same market clearing results.

## 6. Test results

### 6.1. Case study

In order to test the robustness of the proposed mathematical formulation, the proposed market clearing algorithm is evaluated using a test system representing the pan-European electricity market, consisting 42 bidding areas. The aforementioned version of the European transmission grid consists of 59 AC and 13 DC interconnections and is based on the actual interconnections of the European bidding areas, provided to the authors by ENTSO-E, under a non-disclosure agreement (NDA). All ATC values for the continental Europe, the Nord Pool and the six bidding areas of Italy have been taken from publicly available data from the websites of ENTSO-E [43], NordPool Spot [44] and Terna [45]. The topology of the European bidding areas is shown in Fig. 2, where black lines signify the AC interconnections and orange lines signify the DC interconnections.

For demonstration purposes, a set of 274,080 simple hourly priced energy offers (belonging to 1,142 10-step 24-h priced energy offers) are created for all bidding areas, from which 85 are subject to the Minimum Income, Scheduled Stop and Load Gradient Conditions in the Iberian bidding areas (an energy offer may be subject to a combination of the aforementioned conditions), along with 240 priced demand bids for each bidding area. Additionally, a set of 4659 block orders (from which 420 are linked block orders) and 84 flexible hourly orders are also created. All orders are generated based on specific computational formulas, which have been published by the authors.<sup>3</sup>

Finally, ramping related data and initial conditions for all bidding areas and AC/DC transmission lines have been artificially created for demonstration purposes. All data used in the respective case studies, along with the order generation methodology, have been made publicly available by the authors<sup>4</sup> for the efficient reproduction of the herein demonstrated results.

Essentially, what delays the execution process is the number of flow-based constraints, since ATC constraints (without the use of PTDFs) are easy to solve and do not increase the computational requirements of the problem. The number of critical branches that are currently present in the European day-ahead markets (actually in CWE region, which applies the flow-based model) are generally less than 1000 within a trading day (i.e. in August 2015, the average number of the early publication critical branches was 605, whereas the respective minimum and maximum values were 409 and 765, respectively [46]). In our test case (considering only the flow in the AC interconnections in the flow-based constraints (14)), the number of flow-based constraints reaches 2832.<sup>5</sup> Consequently, it is clear that the examined test case constitutes a sound benchmarking test system, for evaluating the effectiveness and computational requirements of the proposed market clearing algorithm.

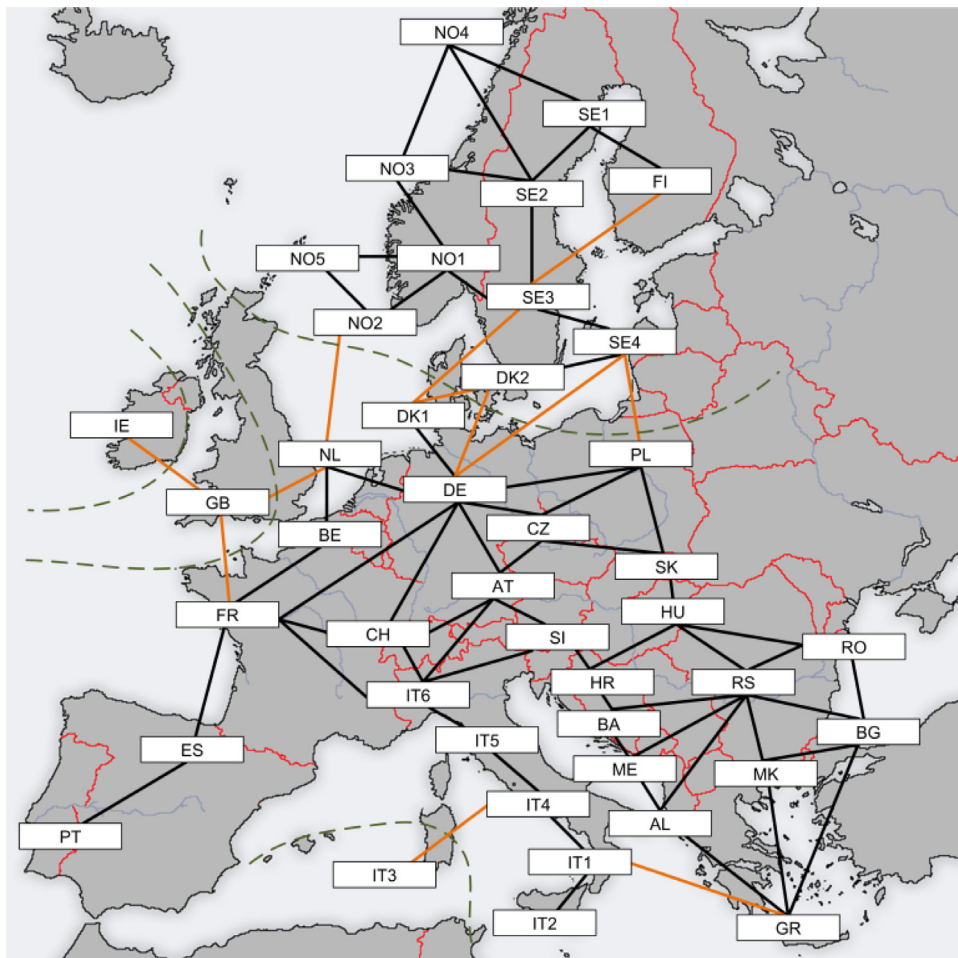
### 6.2. Results

The solution results per iteration of the algorithm are presented in Table 2. As shown, the social welfare decreases in each iteration, due to the fact that paradoxically accepted Blocks and MIC orders are removed from the Order Book, since they do not comply with

<sup>3</sup> <http://emarket.web.auth.gr/site/wp-content/uploads/2015/09/European-Type-Electricity-Market-Clearing-Model-Appendix.pdf>.

<sup>4</sup> <http://emarket.web.auth.gr/site/wp-content/uploads/2015/04/Test-Case.xlsx>.

<sup>5</sup> The total number of flow-based constraints is equal to the number of AC interconnections times the number of the examined trading periods. The aforementioned product should be doubled in order to take into consideration both the maximum and minimum power flow limits. Similarly, the total number of the examined ATC constraints (related to the total number of DC interconnections) is equal to 624.



**Fig. 2.** European power system topology. (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

**Table 2**  
Iterative algorithm results (with activated ramping constraints and losses/tariffs).

| Iteration | Welfare [€]   | PAOs (blocks) | MIC      |         | Execution time [s] |
|-----------|---------------|---------------|----------|---------|--------------------|
|           |               |               | Eligible | Removed |                    |
| 1         | 5,908,080,851 | 12            | 14       | 7       | 43                 |
| 2         | 5,906,902,127 | 5             | 10       | 0       | 35                 |
| 3         | 5,906,901,954 | 2             | 10       | 0       | 38                 |
| 4         | 5,905,058,784 | 1             | 0        | 0       | 36                 |
| 5         | 5,905,058,659 | 0             | 0        | 0       | 37                 |

the respective market rules. Moreover, each iteration of the optimization problem does not require more than 45 s for its successful execution and as a result the overall algorithm terminates in 189 s.

**Table 5** Clearing of various orders during the 1st iteration of the algorithm Columns 3–5 of **Table 2** present the total number of identified paradoxically accepted and eligible/removed MIC orders. It should be noted that in this test case the MIC orders that result in a  $[V_s^{req} - V_s^{mar}]/V_s^{req}$  ratio more than 10% ( $X=10\%$ ) are immediately removed from the Order Book, whereas those with a ratio between 0 and 10% are given another opportunity for clearing (eligible orders), if applicable (initially  $B_s^{mic} = 3$  for all MIC orders). As a result, even though 14 eligible MIC orders have been identified in the first iteration of the algorithm, the number of eligible MIC orders is reduced to 10 in the second iteration, denoting that at least 4 from the aforementioned orders have successfully fulfilled their Minimum Income Condition. The breakdown of the finally attained overall social welfare to its consistent components, namely the profits attained by the respective participants (the profits of supply

**Table 3**  
Social welfare decomposition.

| Participant        | Welfare [€]   |
|--------------------|---------------|
| Supply entities    | 134,089,773   |
| Demand entities    | 5,767,745,663 |
| AC congestion rent | 2,078,131     |
| DC congestion rent | 1,211,516     |
| Tariffs            | –66,424       |
| Total welfare      | 5,905,058,659 |

and demand entities comprise the profit of both simple hourly and block/flexible hourly supply and demand orders), are presented in **Table 3**. As shown, the imposed tariffs in specific DC interconnections result in a decrease in the overall congestion rent (–66,424 €), expressing the additional cost of energy transactions through these interconnections.

**Table 4** provides an overview of the order related results during the final iteration of the algorithm. It should be noted that

**Table 4**  
Overview of order related results.

| Order type             | Cleared (submitted) orders | Cleared (total) quantity [MWh] | % of total cleared quantity | PAO (PRO) | Removed (eligible) MIC orders |
|------------------------|----------------------------|--------------------------------|-----------------------------|-----------|-------------------------------|
| Hourly offers          | 166,295 (248,640)          | 4,979,379 (5,954,051)          | 69.791%                     | –         | –                             |
| Complex offers         | 12,519 (20,400)            | 351,461 (522,846)              | 4.926%                      | –         | 17 (4)                        |
| BO simple <sup>a</sup> | 1895 (3990)                | 1,573,196 (4,032,080)          | 22.050%                     | 16 (22)   | –                             |
| BO linked              | 332 (420)                  | 171,799 (226,939)              | 2.408%                      | 1 (37)    | –                             |
| BO ex. Group           | 78 (249)                   | 56,167 (160,360)               | 0.787%                      | 1 (132)   | –                             |
| FHO supply             | 24 (42)                    | 2671 (4256)                    | 0.037%                      | 1 (0)     | –                             |
| Hourly bids            | 4437 (10,080)              | 7,131,774 (8,635,858)          | 99.985%                     | –         | –                             |
| FHO demand             | 22 (42)                    | 1067 (2657)                    | 0.015%                      | 1 (0)     | –                             |

<sup>a</sup> This block order category contains all regular and profile block orders that are neither linked nor belonging to an exclusive group.

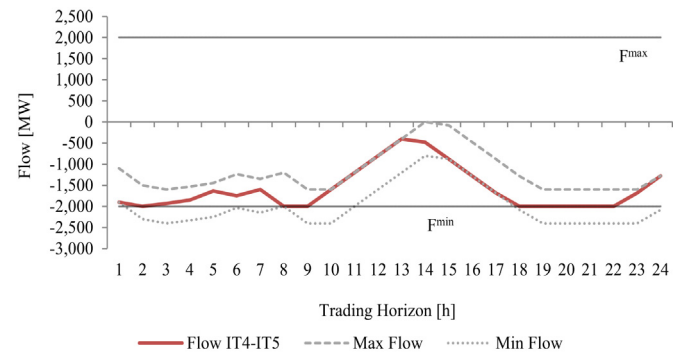
**Table 5**  
Clearing of various orders during the 1st iteration of the algorithm.

| Order  | Parent order | Total welfare [€] | Market welfare [€] | Required welfare [€] | Accepted block order | Paradoxically accepted order | MIC orders |         |
|--------|--------------|-------------------|--------------------|----------------------|----------------------|------------------------------|------------|---------|
|        |              |                   |                    |                      |                      |                              | Eligible   | Removed |
| BO1884 | –            | 10,336.36         | –                  | –                    | Y                    | N                            | –          | –       |
| BO39   | –            | –73.35            | –                  | –                    | Y                    | Y                            | –          | –       |
| BO2275 | –            | –1683.22          | –                  | –                    | Y                    | N                            | –          | –       |
| BO2276 | BO2275       | 2768.13           | –                  | –                    | Y                    | N                            | –          | –       |
| S611   | –            | –                 | 406,154.72         | 426,703.80           | –                    | –                            | Y          | N       |
| S618   | –            | –                 | 427,151.42         | 507,173.12           | –                    | –                            | N          | Y       |

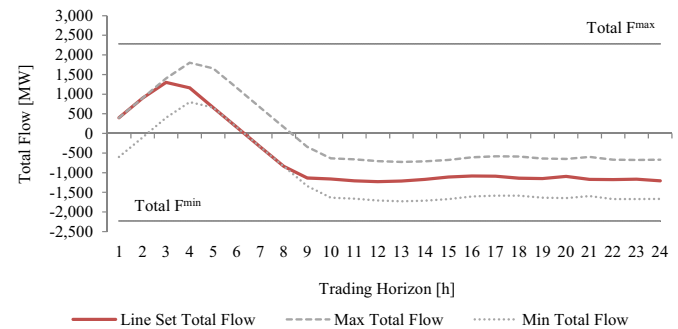
the numbers appearing in the last two columns of Table 4 refer to the total number of Paradoxically Accepted (PAO), Paradoxically Rejected (PRO), Removed and Eligible MIC Orders identified during all iterations of the algorithm. Moreover, the high number of PROs, appearing in the supply block orders belonging to an exclusive group category (BO ex. group), is the result of the incorporation of constraint (10), which prevents a block order to be cleared even though it is “in-the-money”.

Table 5 presents the clearing of various block and complex simple hourly orders during the 1st iteration of the algorithm. The first two rows present the clearing of block orders BO1884 and BO39. As shown, contrary to block order BO1884, which is cleared according to the current market clearing conditions, block order BO39 is paradoxically accepted, since it is cleared, even though it entails a negative welfare to the participant that submits it. As a result, the respective order is removed from the Order Book during the subsequent iterations of the algorithm. The following two rows present the clearing of linked block orders BO2275 (parent block) and BO2276 (child block). The block order BO2275 is accepted in the current market clearing solution, even though it entails a negative welfare equal to –1683.22 €, due to the fact that the welfare of the child block order BO2276 (2768.13 €) is enough to cover the losses of the parent order. Finally, the last two rows of Table 4 present the clearing of two simple hourly orders, which are subject to the Minimum Income Condition. As shown, during the first iteration of the market clearing algorithm simple hourly order S611 is deemed eligible, due to the fact that according to the current Market Clearing Prices, the  $[V_s^{req} - V_s^{mar}] / V_s^{req}$  ratio is equal to 4.8% and thus less than 10% (minimum required ratio for providing another market clearing opportunity). As a result, the respective  $B_s^{mic}$  parameter is decreased to 2 and the aforementioned order is given another opportunity for clearing. On the other hand, the  $[V_s^{req} - V_s^{mar}] / V_s^{req}$  ratio of the simple hourly order S618 is equal to 15.8% and as a result the respective order is removed from the Order Book.

The effect of ramping constraints in the optimization problem solution is illustrated in Figs. 3–6, which present the attained results associated with an individual line, a line set, hourly and daily net position ramping constraints, respectively. In Figs. 3–5 the grey dotted lines denote the upper and lower hourly limits of the power flow and net position variables, which are dynamically calculated, since their value in each trading period depends on the value of



**Fig. 3.** Power flow in the AC interconnection between IT4 and IT5.



**Fig. 4.** Total power flow in the line set containing DC interconnections DK1-SE3, DK1-DK2 and DK1-NO2.

the respective optimization variable during the previous trading period. As shown, the power flow of interconnector IT4-IT5, the total power flow of line set containing interconnectors DK1-SE3, DK1-DK2 and DK1-NO2, and the net position of bidding area SE3 are constrained in some trading periods due to the imposed ramping limits. Moreover, Fig. 6 presents the daily net position results of bidding areas GR, IT2 and PL. As shown, even though the daily net position ramping constraint is not binding for GR, this is not the case for IT2 and PL, which reach their lower and upper daily net position limits (illustrated with black lines in Fig. 6), respectively.

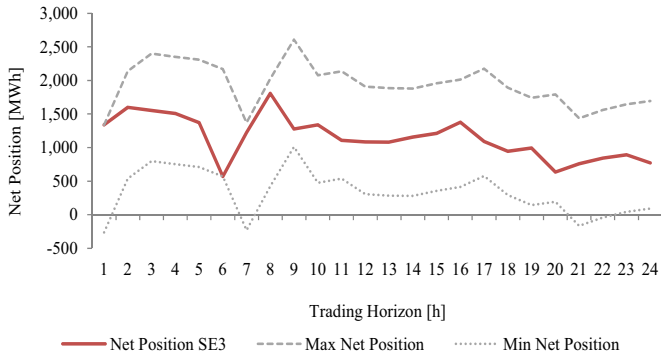


Fig. 5. Net position of bidding area SE3.

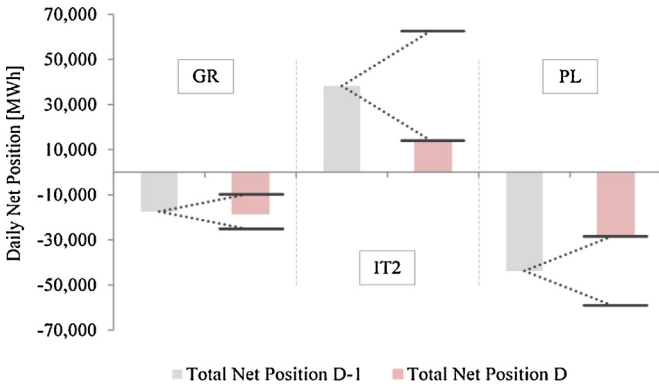


Fig. 6. Daily net position of bidding areas GR, IT2 and PL.

The effect of the net position ramping constraints on the MCPs is illustrated in Fig. 7, where all constituent terms of the finally attained MCPs (Eq. (27)), namely the Lagrange Multipliers associated with the power balance (PB) equation and net position ramping constraints of bidding area FI, are graphically presented. In this figure, columns “RU (H)” and “RD (H)” represent the  $-(\beta_{at}^{H,up} - \beta_{a,t+1}^{H,up})$  and  $\beta_{at}^{H,dn} - \beta_{a,t+1}^{H,dn}$  terms of the finally attained MCPs, respectively, whereas columns “RU (D)” and “RD (D)” represent the  $-\beta_{a,t+1}^{D,up}$  and  $\beta_{a,t+1}^{D,dn}$  terms. Obviously, the final MCPs are equal to the algebraic summation of the aforementioned Lagrange Multipliers.

Fig. 8 and Table 6 present the effect of power flow ramping constraints, tariffs and losses on the MCPs. Fig. 8 presents the power flow, the finally attained MCPs and the Lagrange Multiplier

of the power flow constraint of the DC interconnection between bidding areas DE and SE4. As shown, in trading periods 2–7 the finally attained MCPs do not converge, even though the respective interconnection is uncongested (the respective flow limit Lagrange Multiplier is equal to zero). This is due to the effect of active (binding) ramping constraints, both for the individual line DE-SE4 and for the line set that the aforementioned line belongs to (interconnections DE-SE4 and PL-SE4 belong to the same line set, which is subject to ramping constraints).

Note also that in trading period 3 the power flow between the two aforementioned bidding areas is equal to zero, even though the respective MCPs are not the same. This unexpected result is caused by the imposed tariffs, which have been taken 0.5 €/MWh for the examined DC interconnection. Thus, the MCP difference between the interconnected bidding areas (0.380 €/MWh) does not suffice to cover the tariff for energy transactions through this DC interconnection and the line flow zeroizes.

The first two rows of Table 6 present the Lagrange Multipliers of the power balance constraint of bidding areas DE and SE4, whereas the third row presents the Lagrange Multiplier ( $\mu_{lt} = \mu_{lt}^{max} - \mu_{lt}^{min}$ ) of their interconnection power flow constraint. The final four rows present the Lagrange Multipliers of the interconnection ramping constraints, as follows:

$$\Gamma_{lt}^{H,up} : \gamma_{lt}^{H,up} - \gamma_{l,t+1}^{H,up} \quad (32)$$

$$\Gamma_{lt}^{H,dn} : \gamma_{lt}^{H,dn} - \gamma_{l,t+1}^{H,dn} \quad (33)$$

$$\Gamma_{lt}^{LS,up} : \sum_{ls \in LS} [A_{ls}^l \cdot \gamma_{ls,t}^{LS,up}] - \sum_{ls \in LS} [A_{ls}^l \cdot \gamma_{ls,t+1}^{LS,up}] \quad (34)$$

$$\Gamma_{lt}^{LS,dn} : \sum_{ls \in LS} [A_{ls}^l \cdot \gamma_{ls,t}^{LS,dn}] - \sum_{ls \in LS} [A_{ls}^l \cdot \gamma_{ls,t+1}^{LS,dn}] \quad (35)$$

where parameter  $A_{ls}^l$  is equal to 1 if interconnecting line  $l$  belongs to line set  $ls$  and 0 otherwise. It should be noted that, even though Table 6 presents the values of power balance equation Lagrange Multipliers, these are equal to the attained MCPs, since there are no active net position ramping constraints in both bidding areas. Moreover, the difference between the power balance equation Lagrange Multipliers ( $\xi_{at}, \xi_{a't}$ ) follows the general formula:

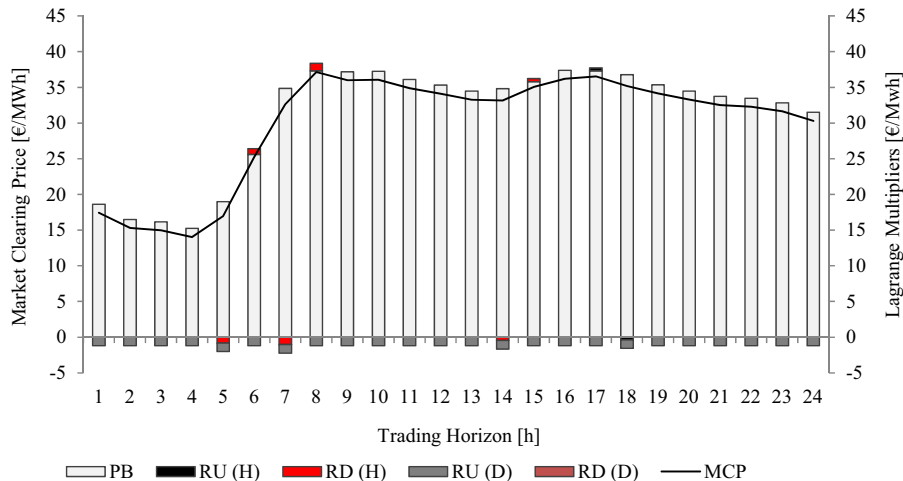


Fig. 7. Market Clearing Prices and Lagrange Multiplier of net position ramping constraints in FI.

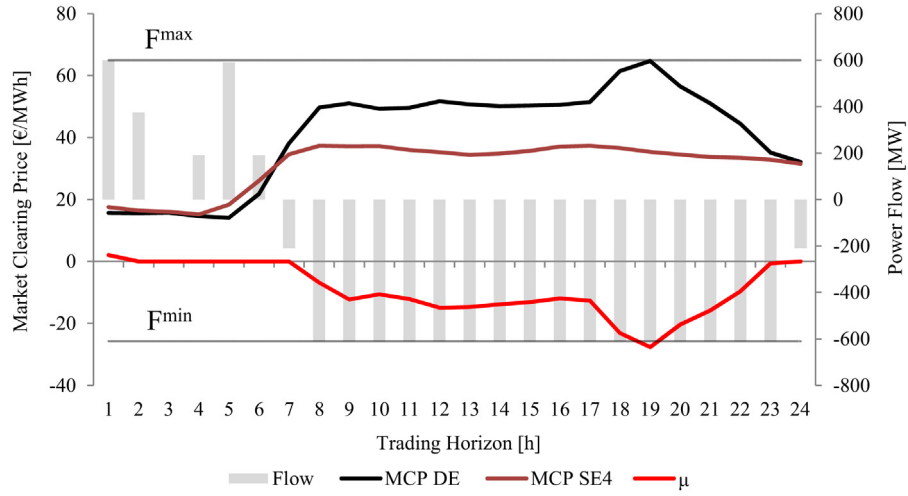


Fig. 8. Market Clearing Prices and Lagrange Multiplier in the DC interconnector between bidding areas DE and SE4.

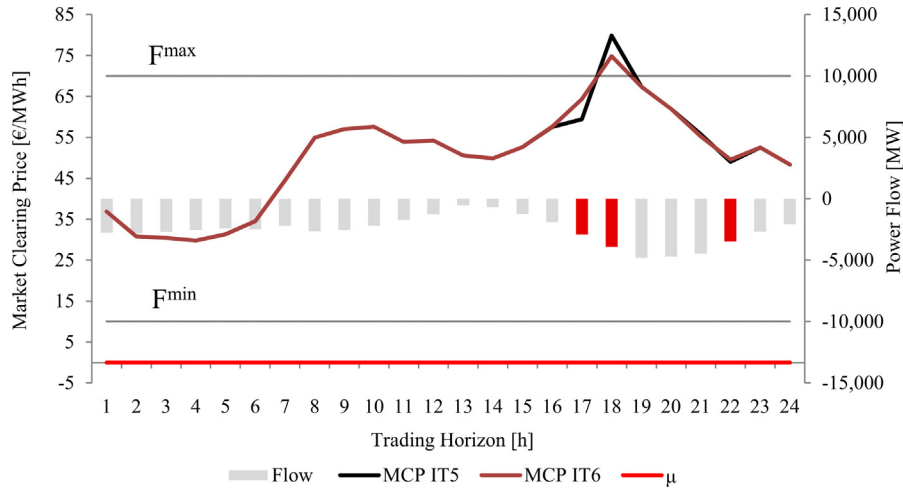


Fig. 9. Market Clearing Prices and Lagrange Multiplier in the radial AC interconnection between bidding areas IT5 and IT6. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 6

Analysis of the MCP difference between areas DE-SE4 in the first eight trading periods.

|                  | Trading Period [h] |               |              |               |               |               |               |               |
|------------------|--------------------|---------------|--------------|---------------|---------------|---------------|---------------|---------------|
|                  | 1                  | 2             | 3            | 4             | 5             | 6             | 7             | 8             |
| $\xi$ DE         | 15.718             | 15.630        | 15.780       | 14.690        | 14.100        | 21.790        | 38.110        | 49.720        |
| $\xi$ SE4        | 18.624             | 16.510        | 16.160       | 15.240        | 18.983        | 25.597        | 34.860        | 37.320        |
| $\mu$            | <b>2.034</b>       | 0.000         | 0.000        | 0.000         | 0.00          | 0.000         | 0.000         | <b>-6.350</b> |
| $\Gamma^{H,up}$  | 0.000              | 0.000         | 0.000        | <b>-0.200</b> | 0.200         | 0.000         | 0.000         | 0.000         |
| $\Gamma^{H,dn}$  | 0.000              | <b>-0.050</b> | <b>0.050</b> | 0.000         | <b>-3.748</b> | <b>-0.753</b> | <b>-0.053</b> | <b>4.555</b>  |
| $\Gamma^{LS,up}$ | 0.000              | 0.000         | 0.000        | <b>-0.055</b> | 0.055         | 0.000         | 0.000         | 0.000         |
| $\Gamma^{LS,dn}$ | 0.000              | 0.000         | 0.000        | 0.000         | 0.000         | <b>-2.042</b> | <b>2.042</b>  | 0.000         |

$$\begin{aligned}
 & \frac{f_{lt}}{|f_{lt}|} \cdot \sum_{a \in A} [(\varphi_l^a - \chi_l^a) \cdot \xi_{at}] - \frac{f_{lt}^+}{|f_{lt}|} \cdot \sum_{a \in A} [\varphi_l^a \cdot \lambda_l^a \cdot \xi_{at}] \\
 & - \frac{f_{lt}^-}{|f_{lt}|} \cdot \sum_{a \in A} [\chi_l^a \cdot \lambda_l^a \cdot \xi_{at}] \\
 & = \frac{f_{lt}}{|f_{lt}|} \cdot (\mu_{lt} + \Gamma_{lt}^{H,up} - \Gamma_{lt}^{H,dn} \\
 & + \Gamma_{lt}^{LS,up} - \Gamma_{lt}^{LS,dn}) + \tau_{lt} \quad \forall l \in L_{dc}, t \in T
 \end{aligned} \quad (36)$$

Eq. (36) highlights the formula linking the finally attained MCPs (power balance equation Lagrange Multipliers) and the imposed tariffs and losses (equal to 0.5 €/MWh and 2%, respectively, for the DE-SE4 interconnection) for all trading periods  $t$  where the interconnector power flow is non-zero.

Similar results can be observed in the case of radial AC interconnections, since both types follow the same modeling principles in the optimization problem. Fig. 9 presents the power flow in the radial AC interconnection between bidding areas IT5 and IT6, the Lagrange multiplier of the power flow constraint and the respective MCPs (both areas are not subject to any net position ramping constraints). As shown, even though the interconnection is not



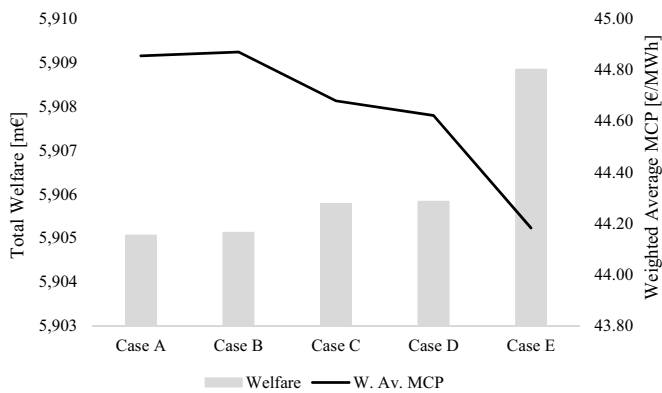


Fig. 10. Attained welfare and average Market Clearing Prices in all examined cases.

congested in any trading period of the trading day (zero flow limit Lagrange multipliers), the MCPs differ in several trading periods, due to the activation of power flow ramping constraints (denoted with red color in Fig. 9), thus leading to price differentiation in the two bidding areas.

Finally, in order to examine the effect of specific constraints on the finally attained results, a sensitivity analysis was performed, where the power flow ramping constraints (Case B), the net position ramping constraints (Case C), the linked block order constraints (Case D) and the complex order clearing conditions (Case E) are deactivated sequentially (Case A contains results associated with the enforcement of all model constraints). The total welfare and weighted average MCPs (with respect to the cleared production) for each examined case are presented in Fig. 10. As shown, the incorporation of complex orders in the market clearing procedure has the greatest effect in the finally attained results, both in terms of social welfare and the resulting market clearing prices.

### 6.3. Computational issues

All cases have been solved in a desktop PC, with an Intel Quad Core i7 CPU processor, running at 3.20 GHz, and 24 GB RAM. The overall mixed integer optimization problem was modeled, implemented and solved in GAMS 24.2.3 modeling environment [47] using the CPLEX 12.6 solver [48]. The relative optimality gap of the MIP solver was set to  $10^{-7}$  for all examined case solutions, while computational times are presented in the last column of Table 1.

It should be noted that in all cases examined by the authors the overall total execution time required for the successful termination of the herein proposed iterative market clearing algorithm was less than 10 minutes, which is the widely accepted (by the European PXs) time frame for the day-ahead electricity market clearing [9].

## 7. Conclusions and further research

In this paper a market clearing model has been presented, which explicitly incorporates the whole range of market orders (except from Merit and PUN orders) and transmission constraints (containing elements of both the ATC and FB network representation) currently appearing in European PXs.

An iterative algorithm for the efficient handling of Paradoxically Accepted Orders and complex MIC conditions has been presented, whereas the proposed model/algorithm has been tested using a test case representative of the magnitude of the real-world European day-ahead electricity market. The attained results revealed important findings about the clearing of market orders and the formation of Market Clearing Prices, especially in the case of active net position and power flow ramping constraints. Such results are extremely useful to market participants and market designers,

since they ensure the efficient operation of the internal European electricity market.

Further research should focus in the incorporation of intra-zonal lines' power flow constraints through the definition and utilization of appropriate Generation Shift Keys, and the efficient handling of PUN orders, currently appearing only in the Italian peninsula. Additionally, the comparison of the herein presented model with other day-ahead market clearing models presented in the literature constitutes a crucial next research step for the authors. Finally, the implications of a switching from the linear pricing model (one price for all settlements) to a non-linear pricing scheme (uplift payments to participants with PAOs) shall be investigated as part of the authors' future research.

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